

Abeer Hashmi

Residence: Hamilton, Ontario *E-mail:* abeerhashmi1212@gmail.com *Number:* 647-594-6165

Education

Bachelor's degree in Engineering Physics

Bachelor's degree program

GPA: 3.7/4.0

McMaster University

Sept 2025 - Present

Engineering Experience

McMaster Mars Rover Team

Mechanical Team Member

September 2026 - Present

Hamilton, Ontario

- Contributed to the development of the V2.75 Mars Rover for CIRC.
- Designed and modified 5+ mechanical components in SolidWorks, applying GD&T and tolerance.
- Performed FEA on mechanical components to evaluate structural performance.
- Managed CAD models, assemblies, and revisions through PDM.

McMaster Engineering Community Outreach

Outreach Facilitator Co-op

May 2026 - August 2026

Hamilton, Ontario

- Delivered 30+ hands-on STEM workshops and engineering demos to K-12 students across the GTA, adapting content for different age groups.
- Planned and facilitated 5+ outreach events to increase STEM engagement among underrepresented communities.

McMaster Pumpkin Chunkin

Power Team Member

December 2025 - August 2026

Hamilton, Ontario

- Modeled a mounting system, chain tensioner, and sprockets in SolidWorks
- Used GD&T and tolerance guidelines to reduce dimensional variation by 15%.
- Produced BOMs and engineering drawings to support manufacturing and assembly.

Apogee Ceramic Inc.

Mechanical Engineering Co-op

September 2024 - February 2025

Paris, Ontario

- Modeled fixtures and safety guards in SolidWorks, applying DFMA to simplify manufacturing.
- Evaluated material and design options using decision-matrices, reducing material cost by 15%.
- Authored and updated 10+ SOPs to align with company policy.

Projects

3D-Printed Warehouse Loading Gripper

Personal Project

2025

Hamilton, Ontario

- Automated loading gripper to transport and handle various shaped objects.
- Applied DFMA to reduce part count by 30% and assembly time by 25%.
- Used decision-matrix analysis to select gripper mechanism, improving strength-to-weight by 15%.
- Iterated 4 prototypes to test and verify grip strength.

Technical skills

CAD & Design

SolidWorks, Autodesk Inventor, Engineering Drawings, BOMs

Mechanical Engineering

GD&T, Decision Matrices, Material Selection, FEA

Programming Languages/Tools

C#, HTML, Python, Matplotlib, NumPy